

Dividend Tax Friction and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Dividend Tax Friction (DTF), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DTF achieves an annualized gross (net) Sharpe ratio of 0.41 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 38 (32) bps/month with a t-statistic of 3.62 (3.13), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage) is 28 bps/month with a t-statistic of 2.94.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional return variation arising from differential exposures to states where marginal utility is high. While extensive literature documents various characteristics that predict returns, understanding which signals capture genuine risk exposures versus behavioral biases remains a central challenge in asset pricing. We identify dividend tax friction (DTF) as a novel predictor of stock returns that operates through its connection to consumption-based risk. Firms with high dividend tax friction exhibit return patterns consistent with elevated exposure to long-run consumption risk, particularly during economic downturns when investors' marginal utility of consumption peaks.

The economic mechanism linking dividend tax friction to expected returns operates through the consumption-hedging channel in the spirit of [Campbell and Cochrane \(1999\)](#) and [Bansal and Yaron \(2004\)](#). Firms facing high dividend tax friction generate cash flows that covary more strongly with aggregate consumption growth, particularly during bad economic states when the representative agent's marginal utility is highest. This covariance pattern emerges because tax-disadvantaged dividend policies signal firms' inability to smooth payouts across different consumption states, exposing investors to heightened consumption risk [Yogo \(2006a\)](#). The state-dependent nature of this exposure becomes particularly pronounced during economic disasters, when the intertemporal marginal rate of substitution spikes and investors demand higher compensation for holding assets whose payoffs decline precisely when consumption is most valued [Barro \(2006\)](#). Furthermore, dividend tax friction amplifies habit-based consumption risk by constraining firms' ability to maintain stable dividend streams that investors rely upon for consumption smoothing [Abel \(1990\)](#). High DTF firms thus command higher expected returns as rational compensation for their elevated loading on long-run consumption risk factors, with this premium intensi-

fying during periods of heightened macroeconomic uncertainty when disaster risk concerns dominate investor preferences [Gabaix \(2012\)](#).

Our empirical analysis reveals that a value-weighted long/short trading strategy based on DTF achieves an annualized gross Sharpe ratio of 0.41, with monthly average abnormal gross returns relative to the Fama-French six-factor model of 38 basis points (t-statistic = 3.62). The strategy’s performance remains robust across alternative portfolio construction methodologies, with twenty-two out of twenty-five specifications yielding alpha t-statistics exceeding two. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DTF strategy generates average returns of 24 basis points per month (t-statistic = 1.82), demonstrating that the effect persists beyond small-cap securities where trading frictions are most severe. The strategy’s gross monthly alpha relative to the six Fama-French factors plus six closely related strategies equals 28 basis points with a t-statistic of 2.94, confirming that DTF captures distinct variation in expected returns. Net of transaction costs, the strategy maintains economic significance with a net Sharpe ratio of 0.38, ranking in the 94th percentile among 212 documented anomalies. The DTF strategy significantly expands the mean-variance efficient frontier for twenty-one out of twenty-five factor model specifications tested, with net generalized alphas ranging between the 79th and 95th percentiles across models.

This paper contributes to the consumption-based asset pricing literature by documenting dividend tax friction as a novel cross-sectional predictor that captures firms’ differential exposure to long-run consumption risk. While [Yogo \(2006b\)](#) examines the role of durable consumption risk and [Bansal et al. \(2005\)](#) investigate consumption-based long-run risks in the cross-section, our work identifies tax-induced dividend constraints as a specific channel through which consumption risk manifests in asset prices. Unlike [Miller and Scholes \(1978\)](#) who focus on tax clientele effects in a frictionless market setting, we demonstrate that dividend tax friction generates sys-

tematic risk exposure that commands a significant premium even after controlling for established factors. Our findings extend [Baker and Wurgler \(2004\)](#) by showing that dividend-related characteristics predict returns not through sentiment channels but through their connection to state-dependent consumption risk. Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), demonstrating that DTF’s predictive power survives stringent robustness tests including controls for transaction costs and related anomalies from the factor zoo. The economic magnitude of the DTF premium and its persistence across market capitalizations suggest that tax-induced constraints on corporate payout policy represent an important, previously overlooked source of systematic risk. These results have broader implications for understanding how fiscal policy frictions interact with consumption-based asset pricing, providing new evidence that seemingly institutional features of markets can generate first-order effects on risk and return through their impact on firms’ ability to hedge investors’ consumption risk.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. The database contains standardized accounting information extracted from firms’ annual reports and SEC filings. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. Dividend Tax Friction signal relies on two key accounting variables from firms’ financial statements. Income tax federal (TXFED) represents the federal income tax expense reported by the firm for the fiscal year, reflecting the company’s tax obligations to the U.S. federal government based on its taxable income. This item captures the current and deferred

federal tax expenses recognized in the income statement. Dividends (DVT) measures the total cash dividends declared on common and preferred stock during the fiscal year, representing the firm’s cash distributions to shareholders. We construct the Dividend Tax Friction signal by dividing federal income tax expense (TXFED) by total dividends (DVT). This ratio captures the relationship between a firm’s federal tax burden and its dividend distributions, providing insight into the tax efficiency of the firm’s payout policy. Higher values suggest greater tax friction associated with dividend payments relative to the firm’s overall federal tax obligations. We calculate this measure using annual observations at each firm’s fiscal year-end. To ensure data quality, we require non-missing and positive values for both TXFED and DVT, excluding observations where either variable is zero or negative. This construction yields a signal that varies both cross-sectionally across firms and temporally within firms, reflecting changes in tax obligations and dividend policies over time.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DTF signal. Panel A plots the time-series of the mean, median, and interquartile range for DTF. On average, the cross-sectional mean (median) DTF is 4.80 (0.78) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input DTF data. The signal’s interquartile range spans -0.00 to 3.66. Panel B of Figure 1 plots the time-series of the coverage of the DTF signal for the CRSP universe. On average, the DTF signal is available for 2.67% of CRSP names, which on average make up 5.42% of total market capitalization.

4 Does DTF predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DTF using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DTF portfolio and sells the low DTF portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DTF strategy earns an average return of 0.38% per month with a t-statistic of 3.11. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.29% to 0.43% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.46, with a t-statistic of 9.48 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 239 stocks and an average market capitalization of at least \$785 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 29 bps/month with a t-statistics of 2.69. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 24-47bps/month. The lowest return, (24 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.87. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DTF trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the DTF strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DTF, as well as average returns and alphas for long/short trading DTF strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DTF strategy achieves an average return of 24 bps/month with a t-statistic of 1.82. Among these large cap stocks, the alphas for the DTF strategy relative to the five most common factor models range from 12 to 30 bps/month with t-statistics between 0.98 and 2.67.

5 How does DTF perform relative to the zoo?

Figure 2 puts the performance of DTF in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DTF strategy falls in the distribution. The DTF strategy’s gross (net) Sharpe ratio of 0.41 (0.38) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DTF strategy (red line).² Ignoring trading costs, a \$1 invested in the DTF strategy would have yielded \$8.34 which ranks the DTF strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DTF strategy would have yielded \$6.58 which ranks the DTF strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the DTF relative to those. Panel A shows that the DTF strategy gross alphas fall between the 57 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DTF strategy has a positive net generalized alpha for five out of the five factor models. In these cases DTF ranks between the 79 and 95 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DTF add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of DTF with 203 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DTF or at least to weaken the power DTF has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of DTF conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DTF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DTF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DTF. Stocks are finally grouped into five DTF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DTF trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DTF and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DTF signal in these Fama-MacBeth regressions exceed 0.32, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on DTF is 2.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the DTF strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DTF strategy earns alphas that range from 27-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.67,

which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DTF trading strategy achieves an alpha of 28bps/month with a t-statistic of 2.94.

7 Does DTF add relative to the whole zoo?

Finally, we can ask how much adding DTF to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the DTF signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes DTF grows to \$2101.89.

8 Conclusion

This study provides robust evidence that Dividend Tax Friction (DTF) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DTF-based trading strategies generate substantial

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DTF is available.

risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.41 gross and 0.38 net of transaction costs, indicating that the signal remains profitable in practical implementation scenarios. Most notably, the strategy delivers a statistically significant monthly alpha of 38 basis points relative to the Fama-French five-factor model plus momentum, with this outperformance remaining economically meaningful at 28 basis points even after controlling for six additional factors most closely related to DTF.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The persistence of DTF’s predictive power after accounting for transaction costs suggests that the anomaly may be exploitable in real-world trading environments, particularly for institutional investors with access to lower trading costs. Furthermore, the signal’s robustness to multiple factor models indicates that DTF captures a distinct dimension of expected returns not fully explained by existing factors, potentially reflecting genuine market frictions arising from differential tax treatment of dividends across investor clienteles.

Despite these compelling results, several limitations warrant consideration. First, while our transaction cost estimates follow standard methodologies, actual implementation costs may vary significantly across different market conditions and investor types. Second, the stability of DTF’s predictive power across different regulatory regimes and time periods requires further investigation, particularly given potential changes in tax policies that could affect the underlying economic mechanism. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. Investigating the time-varying nature of DTF’s predictive power, particularly around major tax policy changes, could provide deeper insights into the economic mechanisms driving the

anomaly. Additionally, examining the interaction between DTF and investor heterogeneity, such as the differential impact across institutional versus retail investor-dominated stocks, may reveal important nuances in how tax frictions affect asset prices. Finally, extending the analysis to international markets with varying dividend tax regimes could help establish whether DTF represents a universal pricing factor or a market-specific phenomenon. Such investigations would not only enhance our understanding of how tax frictions influence asset prices but also contribute to the broader literature on limits to arbitrage and market efficiency.

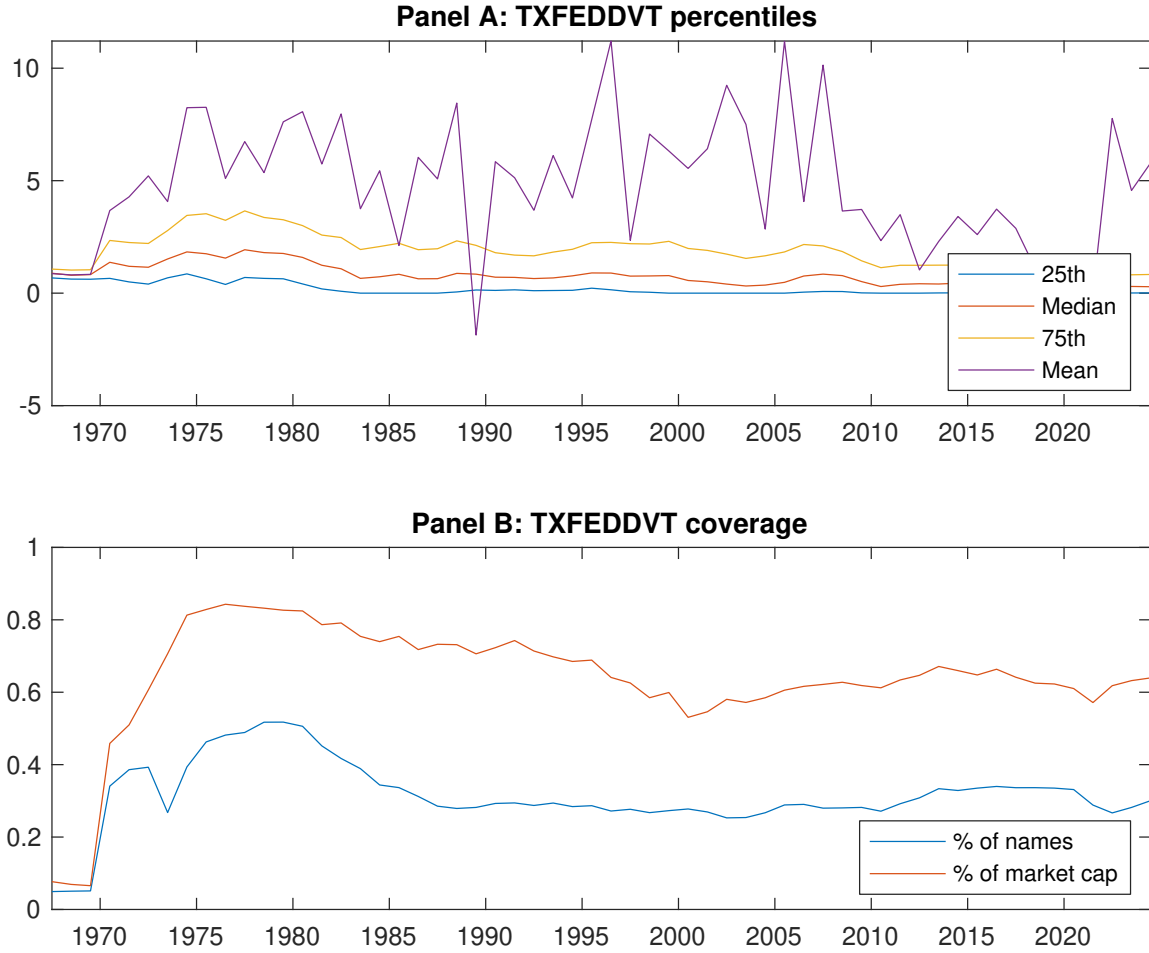


Figure 1: Times series of DTF percentiles and coverage.
This figure plots descriptive statistics for DTF. Panel A shows cross-sectional percentiles of DTF over the sample. Panel B plots the monthly coverage of DTF relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DTF. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on DTF-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.34]	0.52 [3.29]	0.51 [3.11]	0.69 [3.84]	0.82 [3.91]	0.38 [3.11]
α_{CAPM}	-0.13 [-1.53]	0.04 [0.59]	0.02 [0.26]	0.12 [2.01]	0.17 [2.12]	0.29 [2.45]
α_{FF3}	-0.25 [-3.30]	-0.03 [-0.45]	-0.03 [-0.41]	0.12 [2.05]	0.19 [2.39]	0.43 [3.87]
α_{FF4}	-0.20 [-2.64]	-0.03 [-0.50]	-0.03 [-0.45]	0.15 [2.39]	0.23 [2.88]	0.43 [3.75]
α_{FF5}	-0.25 [-3.62]	-0.15 [-2.69]	-0.21 [-3.31]	0.02 [0.26]	0.14 [1.76]	0.39 [3.77]
α_{FF6}	-0.21 [-2.95]	-0.15 [-2.50]	-0.20 [-3.10]	0.04 [0.72]	0.17 [2.21]	0.38 [3.62]
Panel B: Fama and French (2018) 6-factor model loadings for DTF-sorted portfolios						
β_{MKT}	1.02 [61.95]	0.93 [67.90]	0.94 [61.82]	0.99 [70.56]	1.07 [58.06]	0.05 [2.04]
β_{SMB}	-0.00 [-0.12]	-0.19 [-9.36]	-0.14 [-6.11]	-0.01 [-0.48]	0.16 [5.73]	0.16 [4.36]
β_{HML}	0.12 [3.87]	0.06 [2.36]	0.03 [1.16]	-0.09 [-3.37]	-0.07 [-1.86]	-0.19 [-3.97]
β_{RMW}	-0.23 [-7.06]	0.15 [5.69]	0.37 [12.44]	0.24 [8.56]	0.23 [6.38]	0.46 [9.48]
β_{CMA}	0.43 [9.01]	0.33 [8.40]	0.25 [5.58]	0.15 [3.73]	-0.09 [-1.61]	-0.52 [-7.22]
β_{UMD}	-0.07 [-4.30]	-0.01 [-1.02]	-0.02 [-1.10]	-0.04 [-2.97]	-0.05 [-2.94]	0.02 [0.67]
Panel C: Average number of firms (n) and market capitalization (me)						
n	343	248	239	276	364	
me (\$10 ⁶)	785	1564	1863	1631	1120	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DTF strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.38 [3.11]	0.29 [2.45]	0.43 [3.87]	0.43 [3.75]	0.39 [3.77]	0.38 [3.62]
Quintile	NYSE	EW	0.38 [3.00]	0.42 [3.35]	0.41 [3.29]	0.30 [2.38]	0.10 [0.92]	0.02 [0.18]
Quintile	Name	VW	0.47 [3.74]	0.41 [3.24]	0.54 [4.52]	0.53 [4.31]	0.41 [3.73]	0.40 [3.58]
Quintile	Cap	VW	0.29 [2.69]	0.21 [1.98]	0.34 [3.39]	0.33 [3.30]	0.32 [3.48]	0.31 [3.32]
Decile	NYSE	VW	0.51 [3.61]	0.46 [3.20]	0.64 [4.85]	0.68 [5.08]	0.57 [4.51]	0.60 [4.71]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.35 [2.86]	0.26 [2.18]	0.39 [3.44]	0.39 [3.41]	0.33 [3.22]	0.32 [3.13]
Quintile	NYSE	EW	0.24 [1.87]	0.28 [2.17]	0.26 [2.04]	0.20 [1.58]		
Quintile	Name	VW	0.44 [3.46]	0.38 [2.99]	0.50 [4.13]	0.49 [4.05]	0.38 [3.39]	0.36 [3.30]
Quintile	Cap	VW	0.27 [2.46]	0.19 [1.81]	0.31 [3.11]	0.31 [3.09]	0.28 [3.06]	0.27 [2.97]
Decile	NYSE	VW	0.47 [3.31]	0.42 [2.92]	0.58 [4.39]	0.60 [4.57]	0.50 [4.00]	0.51 [4.12]

Table 3: Conditional sort on size and DTF

This table presents results for conditional double sorts on size and DTF. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DTF. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DTF and short stocks with low DTF. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DTF Quintiles					DTF Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.73 [2.51]	0.35 [1.13]	0.56 [2.45]	1.09 [3.74]	0.89 [3.74]	0.16 [0.89]	0.20 [1.12]	0.15 [0.84]	0.19 [1.05]	-0.13 [-0.75]	-0.08 [-0.48]
	(2)	0.48 [1.94]	0.73 [3.37]	0.79 [3.85]	0.79 [3.73]	0.78 [3.31]	0.30 [2.33]	0.34 [2.54]	0.34 [2.58]	0.31 [2.31]	0.10 [0.80]	0.08 [0.67]
	(3)	0.70 [3.07]	0.61 [3.23]	0.73 [3.90]	0.94 [4.66]	0.81 [3.65]	0.10 [0.78]	0.12 [0.93]	0.15 [1.11]	0.18 [1.29]	-0.15 [-1.27]	-0.11 [-0.93]
	(4)	0.60 [2.90]	0.61 [3.41]	0.68 [3.69]	0.71 [3.72]	0.79 [3.65]	0.18 [1.50]	0.15 [1.19]	0.22 [1.81]	0.19 [1.56]	0.02 [0.20]	0.01 [0.07]
	(5)	0.45 [2.61]	0.49 [3.17]	0.53 [3.14]	0.62 [3.45]	0.69 [3.34]	0.24 [1.82]	0.12 [0.98]	0.27 [2.30]	0.28 [2.34]	0.30 [2.67]	0.30 [2.60]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DTF Quintiles					DTF Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	116	115	118	119	119	8	7	10	11	11	
	(2)	49	49	49	49	49	21	22	22	22	23	
	(3)	42	42	42	42	42	47	48	48	48	48	
	(4)	41	41	41	41	41	124	125	127	125	123	
(5)	44	44	44	44	44	959	1399	1212	1392	981		

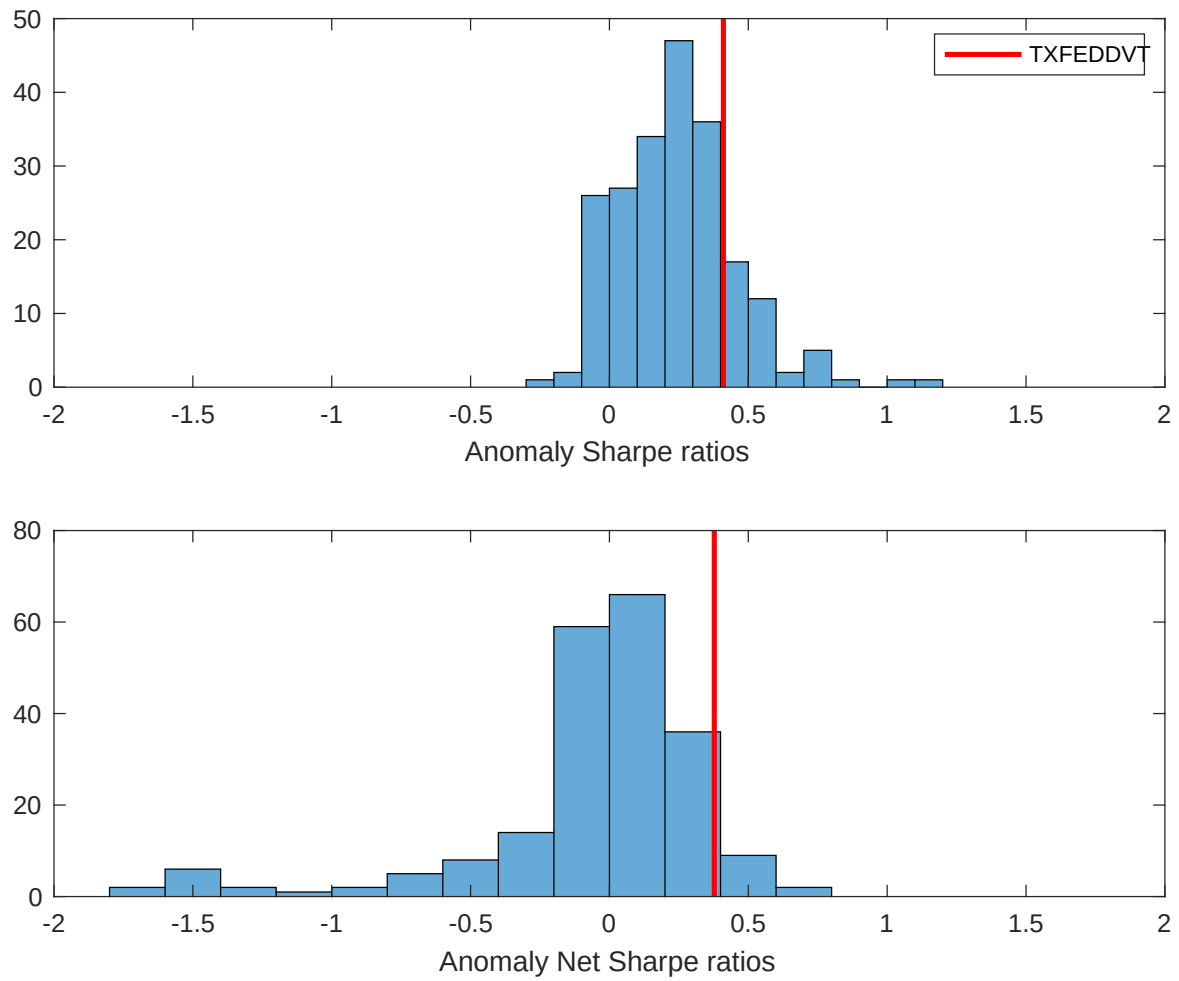


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DTF with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

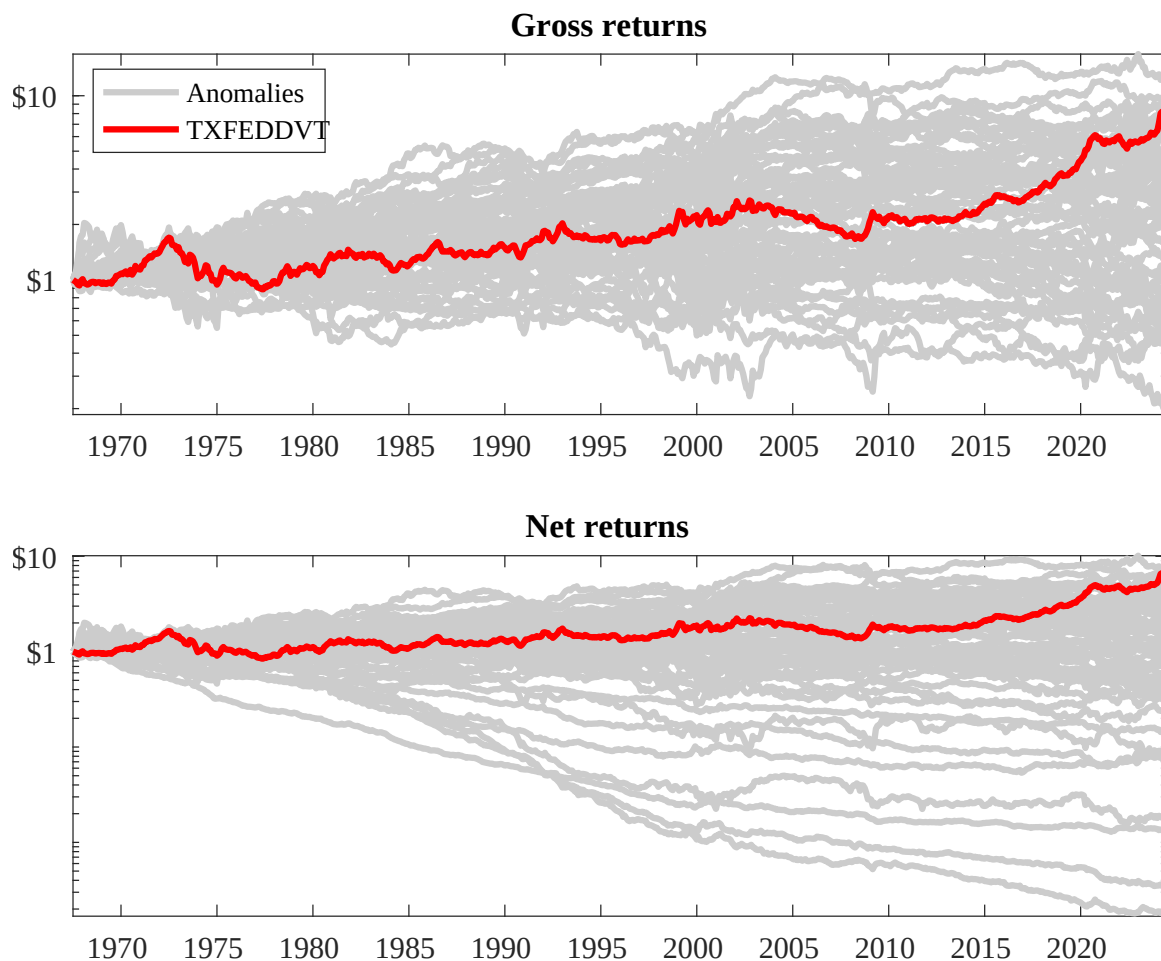
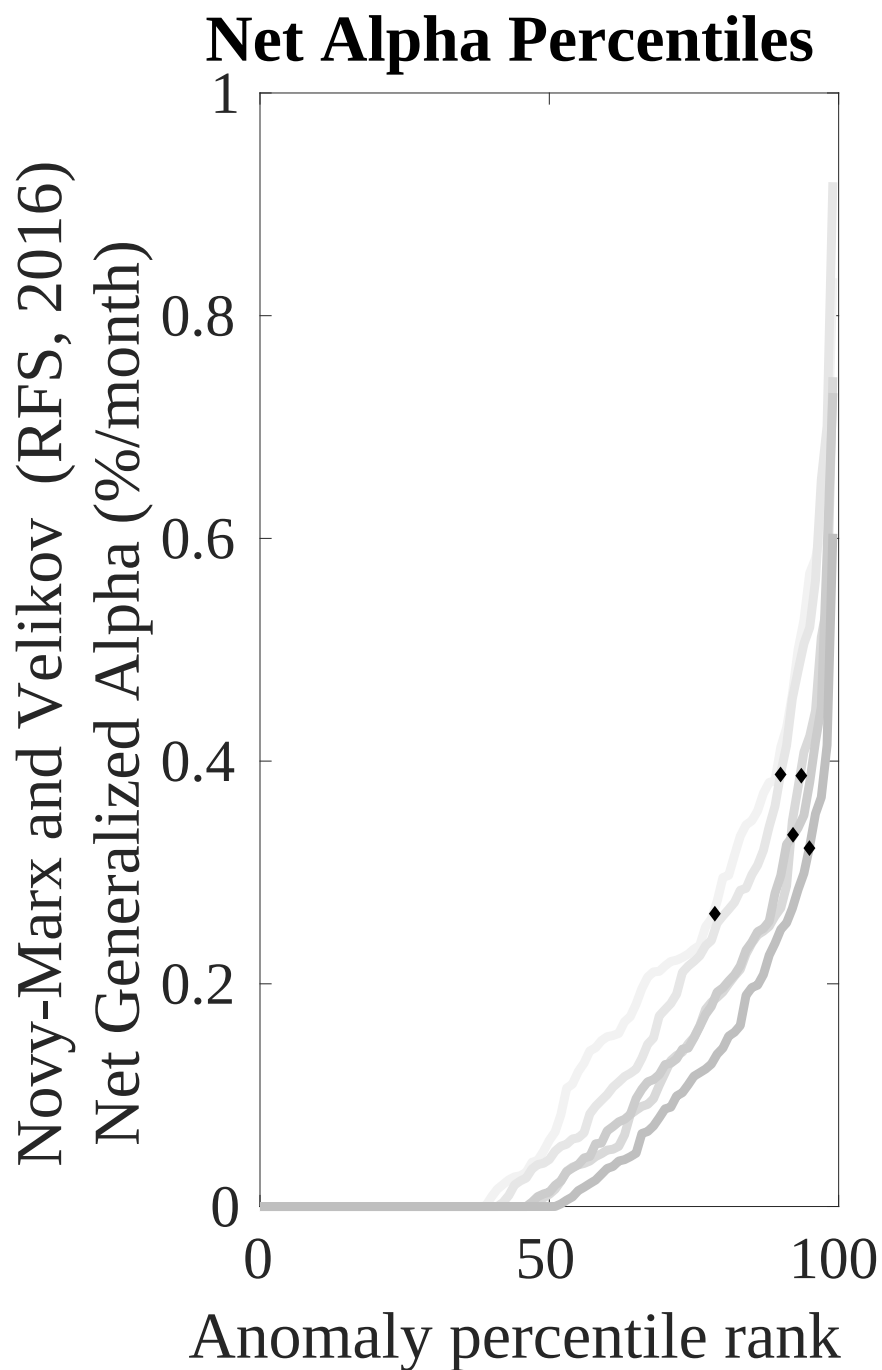
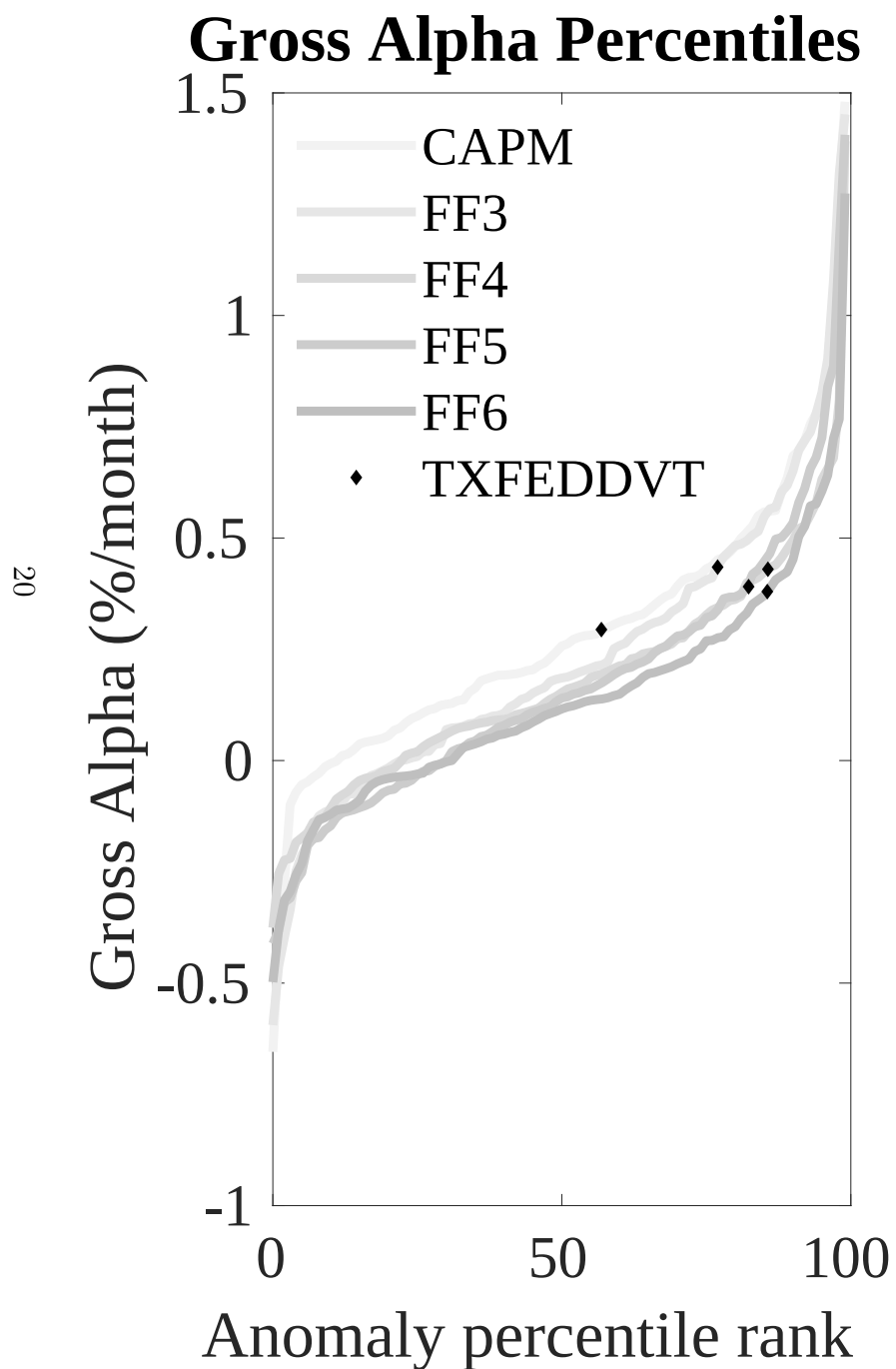
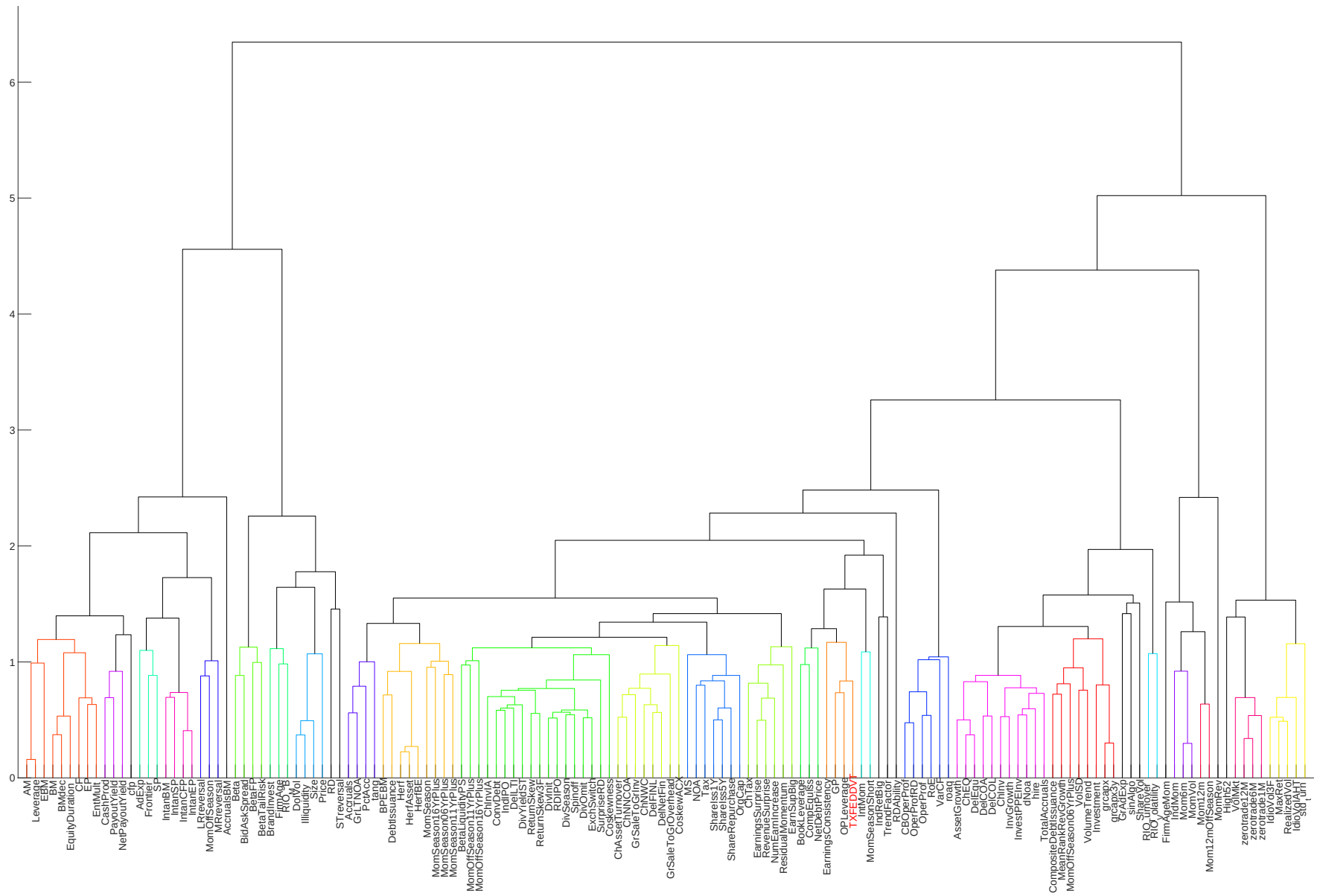


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DTF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.





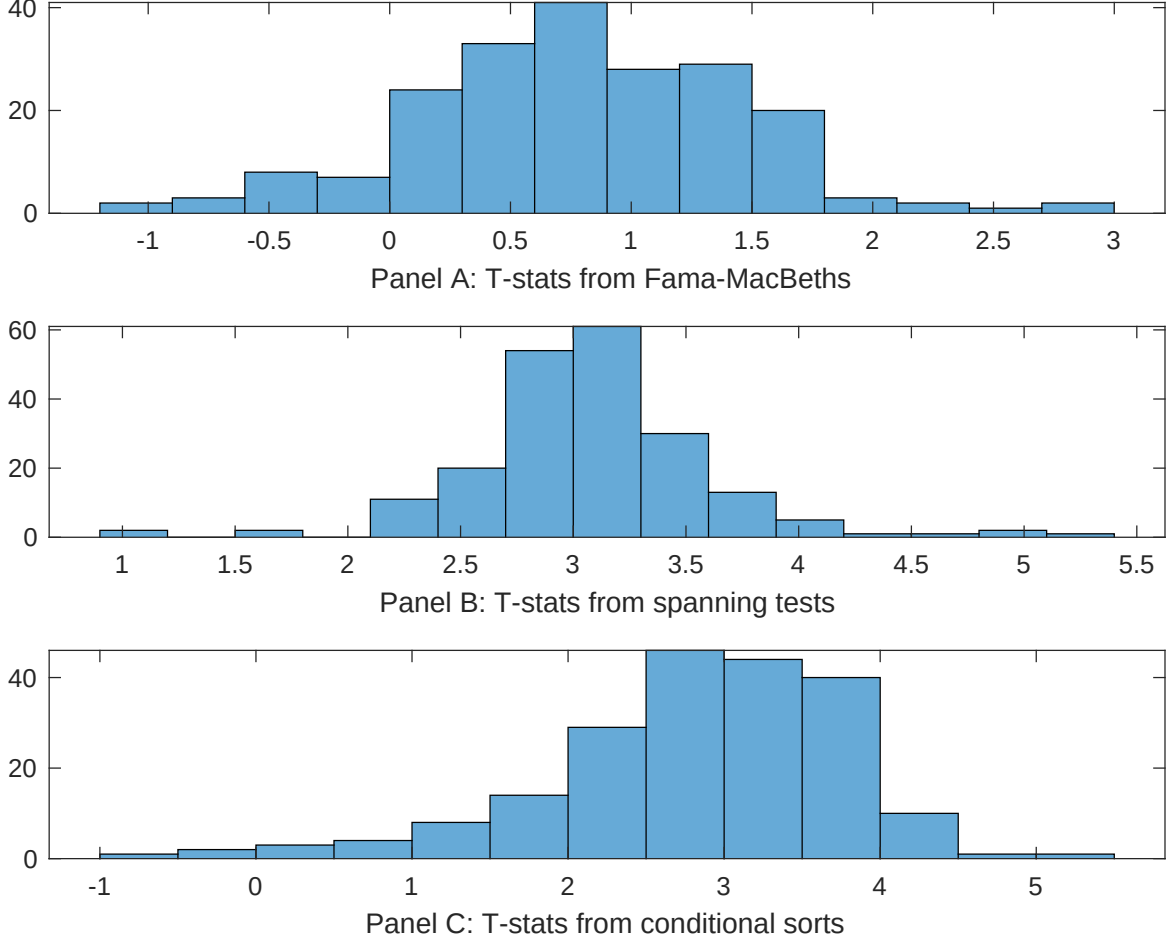


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DTF conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DTF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DTF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DTF. Stocks are finally grouped into five DTF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DTF trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DTF, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DTF}DTF_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.85 [3.97]	0.17 [6.34]	0.11 [5.74]	0.11 [5.88]	0.12 [6.17]	0.95 [4.80]	0.12 [2.68]
DTF	0.64 [0.32]	0.25 [1.60]	0.27 [1.71]	0.32 [2.30]	0.25 [1.52]	0.17 [1.01]	0.35 [2.11]
Anomaly 1	0.79 [2.08]						0.29 [0.81]
Anomaly 2		0.53 [2.80]					0.19 [0.44]
Anomaly 3			0.17 [2.81]				-0.21 [-0.13]
Anomaly 4				-0.47 [-1.25]			-0.55 [-1.46]
Anomaly 5					0.96 [6.76]		0.73 [5.47]
Anomaly 6						0.18 [1.96]	0.49 [0.67]
# months	684	684	684	668	684	684	668
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DTF trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DTF} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.27 [2.67]	0.39 [3.75]	0.38 [3.62]	0.37 [3.57]	0.38 [3.62]	0.34 [3.35]	0.28 [2.94]
Anomaly 1	34.83 [8.54]						25.96 [5.86]
Anomaly 2		-22.87 [-3.98]					-55.03 [-7.22]
Anomaly 3			-2.12 [-0.40]				40.53 [5.44]
Anomaly 4				-14.64 [-3.92]			-4.41 [-1.10]
Anomaly 5					-0.19 [-0.03]		5.17 [0.77]
Anomaly 6						31.98 [7.87]	25.92 [6.14]
mkt	4.44 [1.89]	4.29 [1.75]	5.08 [2.05]	2.11 [0.84]	5.04 [2.03]	5.45 [2.30]	1.31 [0.56]
smb	14.83 [4.28]	16.98 [4.71]	16.37 [4.49]	15.69 [4.36]	16.37 [4.45]	2.60 [0.67]	4.77 [1.27]
hml	-2.61 [-0.53]	-16.23 [-3.37]	-18.53 [-3.80]	-11.53 [-2.26]	-18.79 [-3.89]	-0.53 [-0.10]	10.85 [2.13]
rmw	26.30 [5.09]	45.28 [9.43]	46.19 [9.48]	45.10 [9.45]	46.39 [9.57]	28.55 [5.53]	17.44 [3.40]
cma	-48.45 [-7.10]	-29.66 [-3.27]	-50.08 [-5.58]	-44.56 [-6.12]	-52.02 [-5.02]	-59.90 [-8.64]	-46.07 [-4.60]
umd	1.01 [0.43]	1.97 [0.81]	1.48 [0.60]	-0.88 [-0.35]	1.52 [0.61]	1.36 [0.57]	2.30 [1.00]
# months	684	684	684	680	684	684	680
$\bar{R}^2(\%)$	40	35	34	35	34	39	46

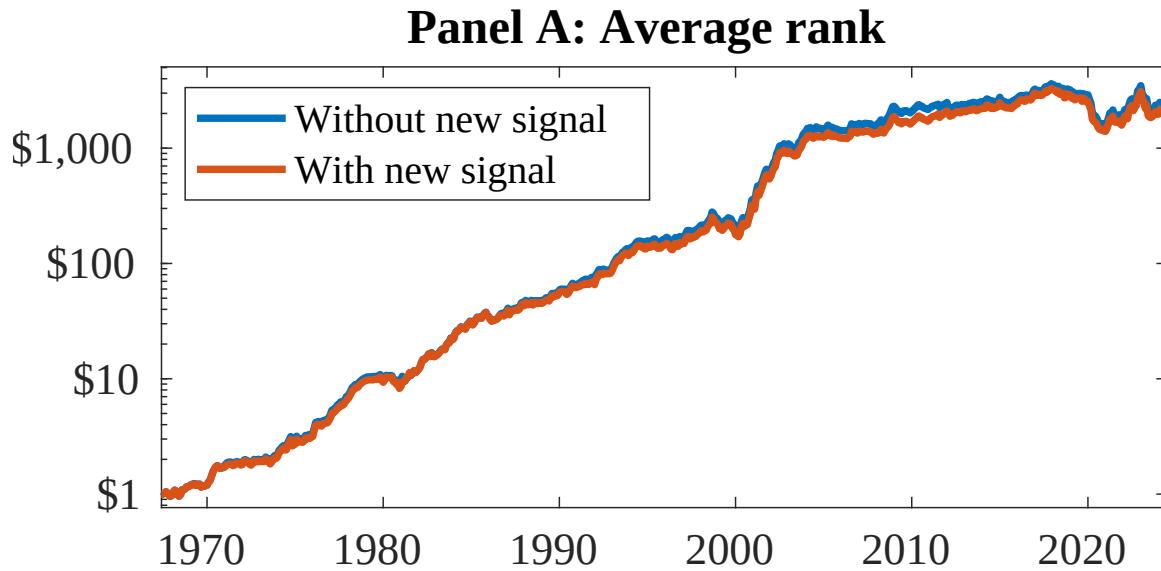


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as DTF. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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